

Name: _____

Date: _____

DIRECTIONS

This assessment covers lessons 4.08.01 to 4.08.07 (entire unit).

Show your work in the box. Write your final answer on the line.

Read every problem carefully. Some problems have more than one part — answer ALL parts.

Take your time. Check your work before you turn in.

1 Unit fraction times whole

Express in simplest form.

(a) $3 \times \frac{1}{9}$ (b) $16 \times \frac{1}{6}$ (c) $7 \times \frac{1}{12}$ (d) $20 \times \frac{1}{4}$

WHOLE $\times$ TOP; KEEP BOTTOM

(a) ____ (b) ____ (c) ____ (d) ____

2 Fraction times whole

Express as a mixed number in simplest form.

(a) $5 \times \frac{5}{6}$ (b) $7 \times \frac{5}{12}$ (c) $8 \times \frac{7}{15}$

MULTIPLY + SIMPLIFY + CONVERT

(a) ____ (b) ____ (c) ____

3 Whole-number or mixed results

Express as a mixed number in simplest form.

(a) $4 \times \frac{3}{4}$ (b) $6 \times \frac{5}{9}$ (c) $15 \times \frac{4}{9}$

MULTIPLY + SIMPLIFY + CONVERT

(a) ____ (b) ____ (c) ____

4 Fraction of a number

Divide by the bottom number; multiply by the top number.

(a) $\frac{4}{5}$ of 15 (b) $\frac{5}{8}$ of 16 (c) $\frac{5}{6}$ of 12

DIVIDE BY BOTTOM; $\times$ TOP

(a) ____ (b) ____ (c) ____

5 More fraction of a number

Same procedure.

(a) $5/9 \times 6$ (b) $3/4 \times 30$ (c) $3/7 \times 18$

DIVIDE BY BOTTOM; $\times$ TOP

(a) ____ (b) ____ (c) ____

6 Fraction of a set

In a basket of 18 apples, 12 are red and the rest are green.

(a) What fraction of the apples are red (simplest form)?

(b) What fraction of the apples are green (simplest form)?

PART OUT OF WHOLE; SIMPLIFY

(a) ____ (b) ____

7 Word problem — multi-part — cat teeth

A house cat has 30 teeth. $\frac{2}{5}$ of the teeth are incisors.

- (a) How many teeth are incisors?
- (b) How many teeth are NOT incisors?
- (c) What fraction of the teeth are NOT incisors (simplest form)?

$\frac{2}{5}$ OF 30; THEN SUBTRACT; THEN FRACTION

(a) ____ (b) ____ (c) ____

8 Word problem — work backward

Asimah's stack of books is $\frac{2}{5}$ nonfiction. She counts 30 nonfiction books.

- (a) Using a bar model, how many books equal $\frac{1}{5}$ of the stack?
- (b) How many books are in the whole stack?

BAR MODEL: 2 PARTS = 30

(a) ____ (b) ____

9 Word problem — pool

The community pool holds 48,000 gallons when full. It is currently $\frac{3}{4}$ full.

(a) How many gallons are in the pool now?

(b) How many more gallons would fill it?

$\frac{3}{4}$ OF 48,000; THEN SUBTRACT

(a) ____ (b) ____

10 FIND THE MISTAKE

A student wrote: " $6 \times \frac{5}{9} = \frac{30}{54}$ " and stopped there.

(a) What did he forget to do?

(b) Show the correct simplest-form answer (as a mixed number).

NAME ERROR; FIX

(a) ____ (b) ____

11 EXPLAIN — Unit 8 reflection

You learned three big moves: multiply a fraction by a whole, find a fraction of a number, and work backward from a fractional part.

Pick ONE move and describe a real situation where you would use it.

WRITE YOUR REFLECTION

My explanation: